

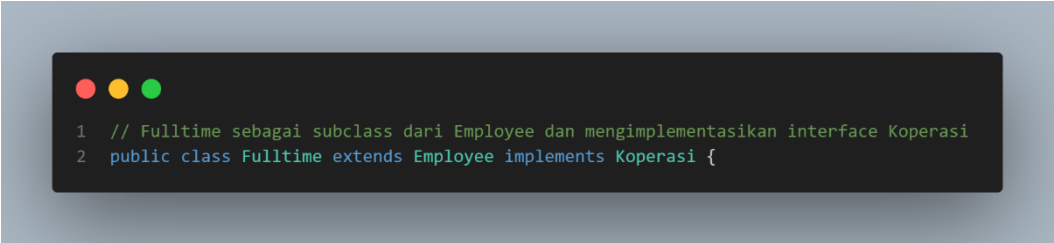
JAWABAN UTS

Nama: Fauzan Juniar Mulyana

NIM: 241524011

BAGIAN 1 A

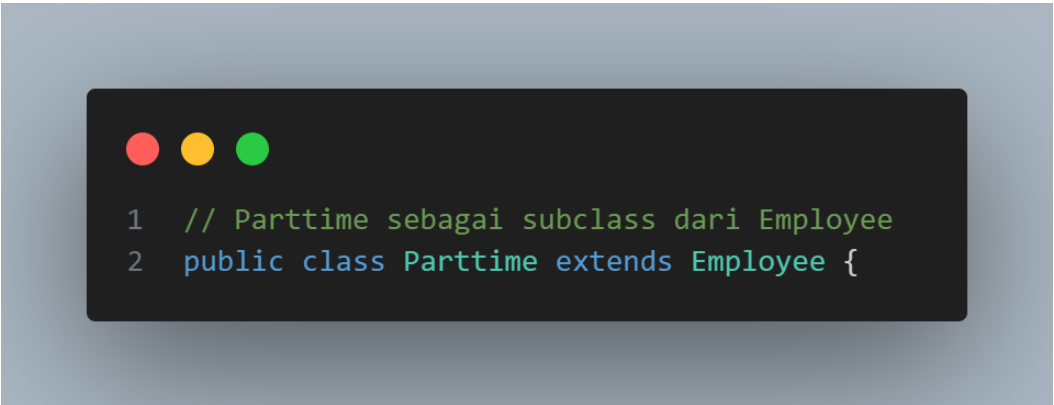
A. Penerapan Multiple inheritance



```
1 // Fulltime sebagai subclass dari Employee dan mengimplementasikan interface Koperasi
2 public class Fulltime extends Employee implements Koperasi {
```

Class Fulltime melakukan extends ke abstract class Employee sehingga, class Fulltime merupakan subclass dari class Employee, sedangkan abstract class Employee adalah superclass. Kemudian, class fulltime juga mengimplementasikan interface dari Koperasi

B. Penerapan Multiple inheritance (2)



```
1 // Parttime sebagai subclass dari Employee
2 public class Parttime extends Employee {
```

Class Parttime melakukan extends ke abstract class Employee sehingga, class Fulltime merupakan subclass dari class Employee, sedangkan abstract class Employee adalah superclass.

Bagian 2 A

```
1 // Employee sebagai superclass dari Fulltime dan Parttime
2 public abstract class Employee {
3     // Membuat instance variable
4     protected String salary;
5     protected department departments;
6     // Membuat Constructor
7     // Menggunakan class department sebagai parameter menerapkan has-a relationship
8     public Employee(String salary, department departments) {
9         this.salary = salary;
10        this.departments = departments;
11    }
12
13    // Getter untuk salary
14    // Method untuk mendapatkan gaji
15    public abstract String getSalary();
16 }
17
18 // class department Sebagai class untuk menyimpan nama departemen
19 class department {
20     // Membuat instance variable
21     private String name;
22
23     // Constructor untuk kelas department
24     public department(String name) {
25         this.name = name;
26     }
27
28     // Method untuk mendapatkan nama departement
29     public String departmentName() {
30         return name;
31     }
32 }
```

Kelas yang menerapkan agregasi *has-a relationship* adalah class Employee dan juga class Department, source code di atas menunjukkan bahwa class Employee menggunakan class department sebagai parameter yang menunjukkan bahwa Employee has-a departments

Bagian B

Employee.java

```
1  import java.time.LocalDate;
2  import java.util.ArrayList;
3  import java.util.List;
4
5  public abstract class Employee {
6      protected String name;
7      protected String id;
8      protected LocalDate joinDate;
9      protected String position;
10     protected double basicSalary;
11     protected List<Lembur> overtimeList;
12     protected int numberOfChildren;
13     protected Department department;
14
15     public Employee(String name, String id, LocalDate joinDate, String position, double basicSalary, Department department) {
16         this.name = name;
17         this.id = id;
18         this.joinDate = joinDate;
19         this.position = position;
20         this.basicSalary = basicSalary;
21         this.department = department;
22         this.overtimeList = new ArrayList<>();
23         this.numberOfChildren = 0;
24     }
25
26     public void addOvertime(Lembur overtime) {
27         overtimeList.add(overtime);
28     }
29
30     public void setNumberOfChildren(int numberOfChildren) {
31         this.numberOfChildren = Math.min(numberOfChildren, 3); // Maximum 3 children
32     }
33
34     public abstract double calculateSalary();
35
36     // Getters
37     public String getName() { return name; }
38     public String getId() { return id; }
39     public LocalDate getJoinDate() { return joinDate; }
40     public String getPosition() { return position; }
41     public double getBasicSalary() { return basicSalary; }
42     public List<Lembur> getOvertimeList() { return overtimeList; }
43     public int getNumberOfChildren() { return numberOfChildren; }
44     public Department getDepartment() { return department; }
45     public String getDepartmentName() { return department.getDepartmentName(); }
46 }
```



```
1  public class Department {
2      private String departmentName;
3      private String departmentId;
4
5      public Department(String departmentName, String departmentId) {
6          this.departmentName = departmentName;
7          this.departmentId = departmentId;
8      }
9
10     public String getDepartmentName() {
11         return departmentName;
12     }
13
14     public void setDepartmentName(String departmentName) {
15         this.departmentName = departmentName;
16     }
17
18     public String getDepartmentId() {
19         return departmentId;
20     }
21
22     public void setDepartmentId(String departmentId) {
23         this.departmentId = departmentId;
24     }
25 }
```